Evaporation

Final Lab Report
Unit Operations Lab 4
October 27, 2025

Group 3
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1. Executive Summary (Examples at end of the Guideline) (10 pts)

The objective of this experiment is to evaluate how differing vacuum pressure affects the performance of a single-effect evaporator. By varying vacuum pressures, the economy and capacity are calculated and compared by using a 10 wt% sugar solution. The significance of the experiment lies in understanding how pressure affects the boiling point and overall heat transfer, and therefore the efficiency of the evaporator. The experiment was done at absolute vacuum pressures of 125, 379, and 506 torr/mmHg. Results show that as absolute pressure decreased, the economy and capacity increased. The highest capacity was 5 kg/hr at an absolute pressure of 125 torr/mmHg, while the lowest capacity was 1.72 kg/hr at an absolute pressure of 506 torr/mmHg. The highest economy was 0.454 kg product/kg steam at an absolute pressure of 125 torr/mmHg, while the lowest economy was 0.19 kg product/kg steam at an absolute pressure of 506 torr/mmHg. The findings suggest that lowering the vacuum pressure significantly improves both the capacity and steam economy of the system, making it more efficient for industrial applications where energy consumption is a critical factor. However, the trade-off includes higher energy losses due to the increased steam required at lower pressures. [1]

2. Introduction (10 pts)

Evaporation is a core process in various industries, including food and beverage, pharmaceuticals, and water treatment industries for concentrating solutions by removing a volatile solvent, typically water. In the food and beverage industry, evaporators are used to concentrate products to improve shelf life and reduce transportation costs. In pharmaceuticals, they intensify active ingredients and enable solvent recovery during drug manufacture. The process involves heating the solvent, leaving a more concentrated solute steam. Evaporator performance is commonly characterized by the capacity, or mass of solvent evaporated per hour, and the steam economy, or the ratio of solvent evaporated to steam consumed. Optimizing these parameters is critical for energy-intensive operations where utility costs dominate. The objective of this experiment was to investigate how varying vacuum pressures influence the performance of a single-effect evaporator processing a 10 wt% sugar solution. to investigate how varying vacuum pressures influence the performance of a single-effect. The evaporator was operated at three absolute vacuum pressures— 10 inHg, 15 inHg, and 25 inHg. This was to observe how pressure changes can influence capacity and economy. Lower absolute vacuum pressures reduce the boiling point of the solution and increase the temperature difference between the steam and the solution. This can enhance the heat transfer rate and improve the overall system capacity. However, the effect of a lower pressure on the economic performance of the operation can vary, as it might increase steam consumption. The experimental setup included a single-effect evaporator with a vacuum pump, heat exchanger, condenser, steam trap, and thermocouples for temperature monitoring. The 10 wt% sugar solution was maintained at a constant concentration throughout the experiment by adding makeup water as needed. Steam was utilized to heat the solution, causing evaporation, and the condensed water vapor was collected and measured to calculate the system’s capacity. The economy was calculated by comparing the mass of the evaporated water to the steam consumed during the process. The experiment was performed under three vacuum pressure settings to determine the optimal pressure for maximizing both the capacity and the economy. The results of this experiment are significant for industries that rely on evaporation for liquid concentration. Gaining an understanding of how differing vacuum pressures affect evaporator performance can lead to more energy-efficient designs and processes, reducing energy consumption and operational costs. [2, 3]

3. Theory (10 pts)

Evaporation is used to separate a concentrated solution via thermal separation. In this experiment, a single-effect evaporator is used to evaporate the sugar water solution at various vacuum pressures to study how different pressures affect boiling point, heat transfer, and the overall performance of separation. The evaporator acts as a heat exchanger where steam condenses on one end and transfers the heat to the boiling solution. Here, the driving force of heat transfer is the temperature difference between the condensing stream and the boiling solution. The results to be obtained are the following: capacity and economy, and to do this, an energy balance around the evaporator must be done while considering the following design equation.

$$q = UA\Delta T$$

Were the following variables defined

U = the overall heat transfer coefficient
A = the area available for the heat to transfer
ΔT = the average temperature difference (diff between condensate and boiling fluids)

To calculate the capacity, we take the amount of product in kg (condensate) collected and divide it by the amount of time in hours it took to calculate. And to calculate the economy, we divide the amount of kg of product by the amount of kg of steam used.

The 2 main values of interest are the capacity and economy. These are both measurements of the performance of the evaporator at various parameters. The capacity is the number of kilograms of water vaporized by the steam fed to the unit. The economy is the ratio,

$$\frac{kg\, of\ product}{kg\, of\ steam\, used}$$

In a single-effect evaporator, the economy is always less than 1, but in a multi-system it may be considerably higher. The steam consumption is always in kg/hour which is equal to, [4]

$$\frac{capacity}{economy}$$

4. Results (15 pts)

Capacity Correlations

Capacity- Kilograms of product collected per unit time (hours in this case)

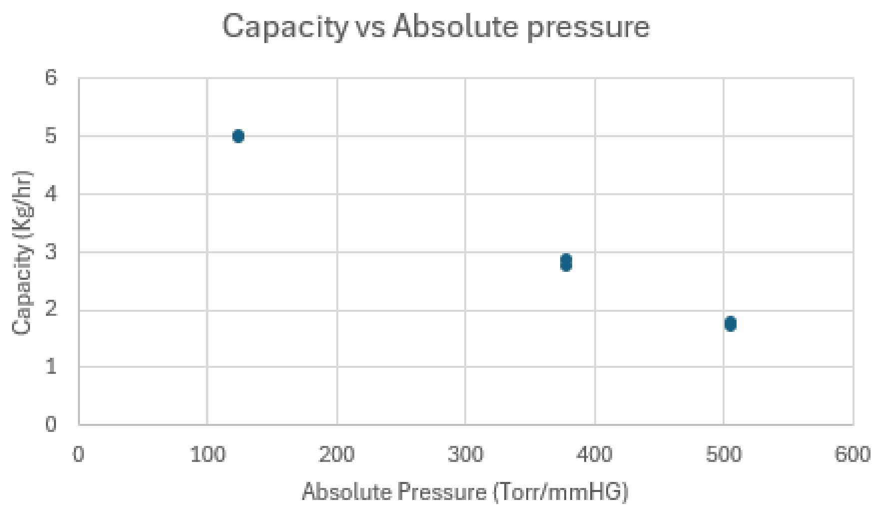

Figure 1: Capacity (kg/hr) vs Absolute pressure (torr/mmHg)

The above graph was obtained by plotting the capacity values at different absolute pressures, which were the following (125, 379, 506) Torr/mmHg. From the graph, we notice that when the pressure increases, approaching atmospheric pressure, the capacity decreases. This trend aligns with the principle of evaporation, where increasing the vacuum pressure lowers the boiling point, enhancing the driving force for heat transfer; therefore, operating at lower pressure is more efficient for evaporation, giving us a higher capacity.

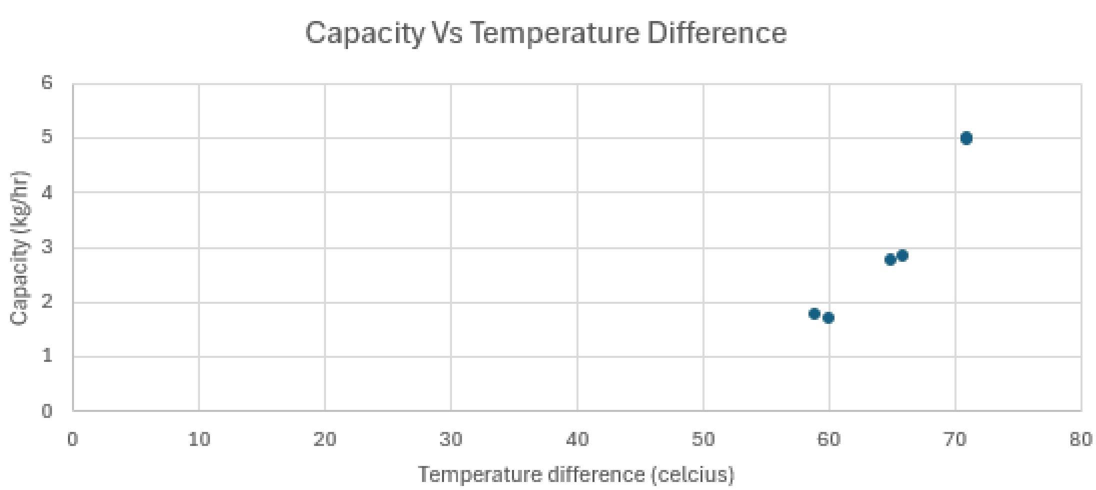

Figure 2: Capacity (kg/hr) vs Temperature Difference (C)

The above graph shows the relationship between the temperature differences between the steam and the solution at each vacuum setting. As the temperature difference increases, the capacity also increases. This supports the trend that a larger temperature gradient is more effective for heat transfer, increasing the evaporation rate.

<u>Economy correlations</u>

The economy in this lab is the ratio between the amount of vaporized product to the amount of steam used. A higher economy would naturally indicate better use of the steam used.

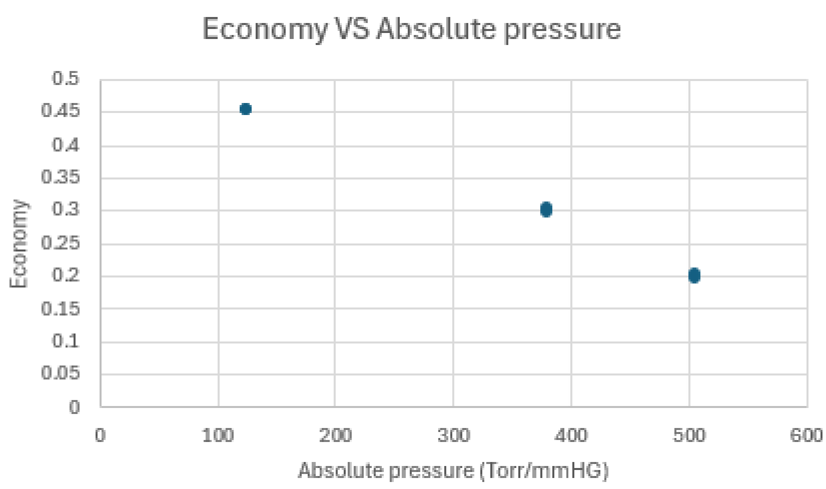

Figure 3: Economy vs Absolute Pressure (torr/mmHg)

The above graph shows the relationship between the Economy and the Absolute pressure. As the absolute pressure increases, the boiling point increases, reducing the temperature difference between the steam and liquid, which correlates with poor heat transfer efficiency. In conclusion, the economy decreases with increasing absolute pressure because the system becomes less efficient at using the steam to vaporize the liquid at higher pressures.

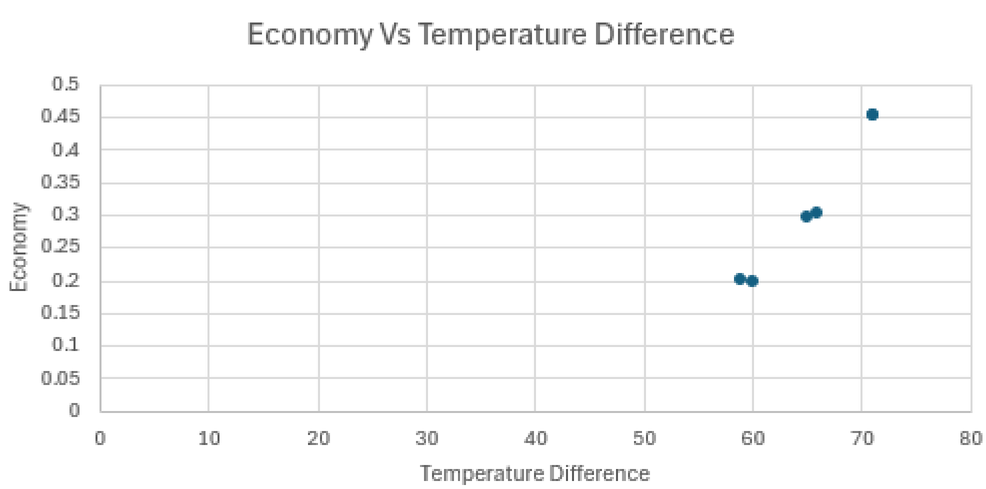

Figure 4: Economy vs Temperature Difference (C)

The above graph shows the relationship between the economy and the Temperature difference. This graph further reinforces that smaller temperature differences result in a poor economy because there is less heat transfer driving force, causing a higher dependence on steam, therefore resulting in a poor economy.

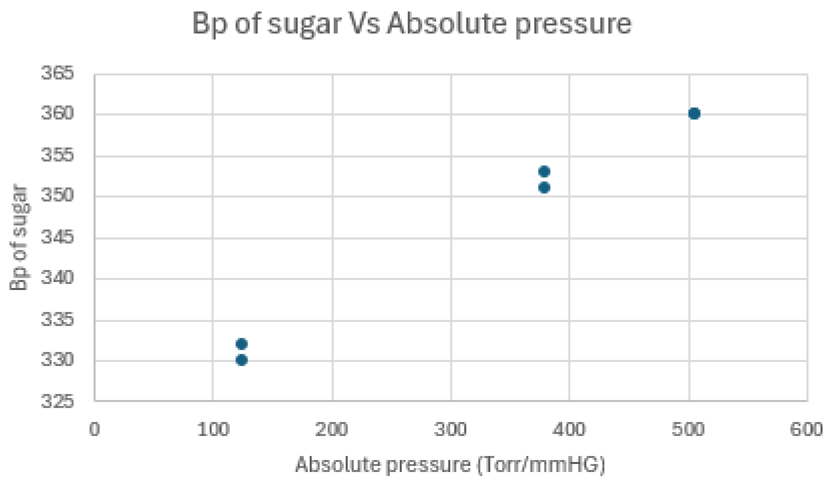

Figure 5: Boiling Point of Sugar Solution (C) vs Absolute Pressure (torr/mmHg)

The above graph shows the relationship between the Boiling point of sugar vs absolute pressure. As previously stated, lower Absolute pressures result in lower boiling points of the solution, and if the absolute pressure increases, so does the boiling point of the solution, and the data shows that as higher pressures require more energy to start boiling the solution

Energy Balance

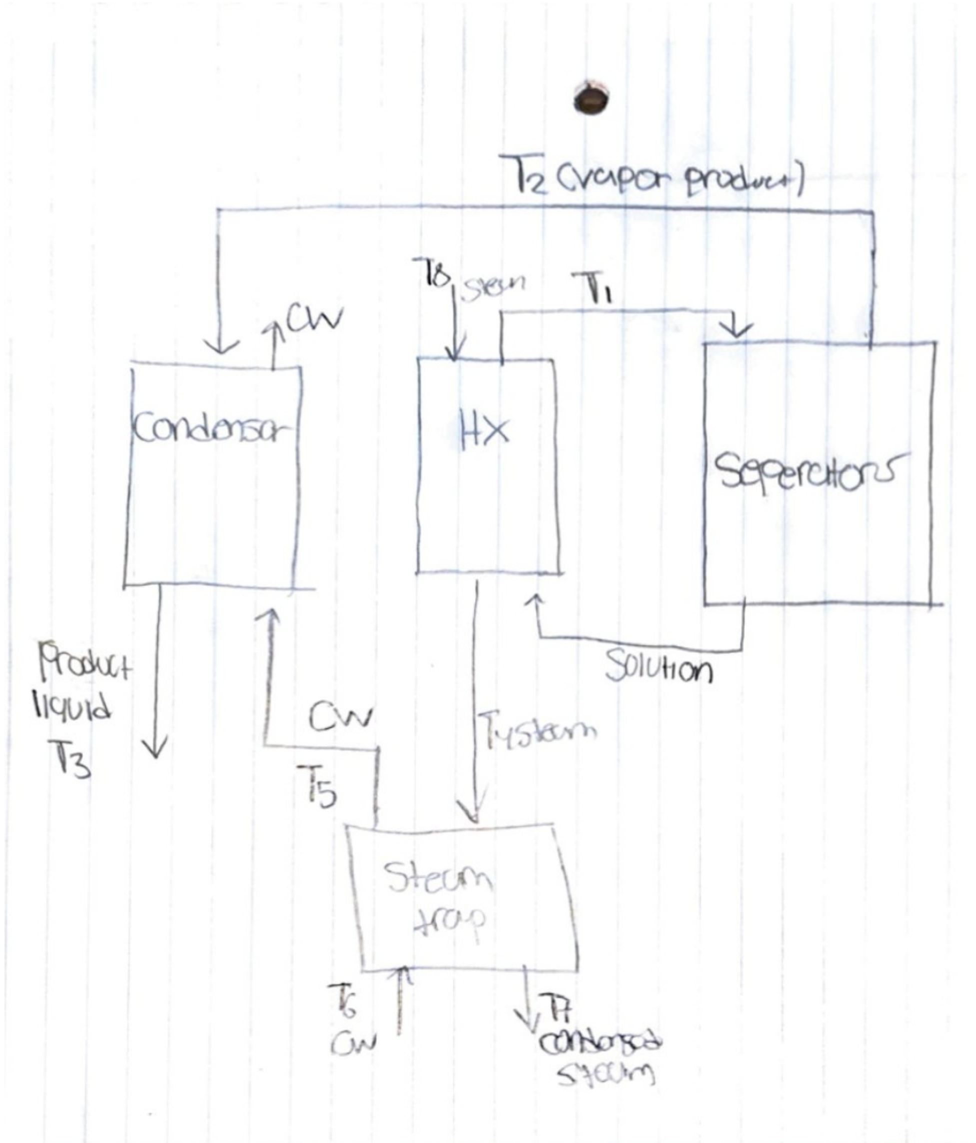

Figure 6: Energy Balance Diagram

5. Discussion (15 pts)

The goal of this experiment was to determine the correlation between vacuum pressure and the performance of the evaporators. It was viewed that changing the pressure decreases the boiling point and allows for less heat to be supplied, with less steam supplied, a better economy can be obtained. In this experiment, we had pressure values of 10inHg, 15inHg, and 25inHg.

The following paragraphs are responses to the “Further Questions to Answer” section in the lab manual.

DI water with no impurities would be the makeup solution to keep a steady sugar concentration throughout the experiment. Using this solution eliminates the effects of sugar concentration on evaporation ratio or efficiency; the evaporation itself gets affected depending on the level of the vacuum pressure. Conversely, a diluted solution gets continuously concentrated and taken out of the system in an industrial setup.

The heat of vaporization for steam is a function of pressure. Which means that a higher value of pressure results in a lower value of latent heat. The energy of the vapor and liquid phases is similar, so the heat of vaporization can be determined by using a steam table by finding the difference between the saturated vapor energy and saturated liquid energy at a specific value of temperature and pressure.

Other than vacuum pressure, several factors affect evaporator performance. The higher sugar concentration increases viscosity and elevates boiling points, reducing heat-transfer efficiency. The flow rate of steam and the effects of pressure influence the temperature driving force. The temperature of the feed and the cooling water conditions also alter the evaporation system’s heat balance and can change both the capacity and economy of the evaporator itself.

As for the discussion of the results section, the experimental findings tend to agree with the theoretical predictions. As the vacuum pressure was increased (reduced value of absolute pressure), the boiling point of the sugar solution lowered, raising the difference in temperatures between the steam and the sugar solution. As a result, there was an enhancement in the evaporator’s capacity, which is in line with theoretical concepts of single-effect evaporators. Deviations from perfect experimental values could be attributed to losses of heat in experiments.

For the experimental data given, the data were recorded over a 15-minute period. A few key takeaways from the calculations are the correlation between pressure and steam economy. As the pressure increased, the economy decreased, showing an inverse relationship between pressure and economy. Moreover, the same can be said for the capacity, showing that economy and capacity have a linear relationship when compared to pressure.

A higher heating rate was observed from the steam going into the heat exchangers as the evaporating pressures decreased. This shows that there was less heat required for the evaporation of the solvent from the sugar-water mixture since there was more heat required by the steam to be condensed after exiting from the heating exchanger. This shows an efficient process as we have managed better capacity using less heat. Efficiency can also be seen by the fact that time was kept unchanged for all experiments, meaning time was not an independent variable, which could increase or decrease the process. The temperature will, therefore, depend on the ratio of the mass/matter of the product itself to the mass of steam used in producing the product (the economy).

The most likely sources of error would be the incorrect measuring of the volume of liquids using the graduated cylinders, variations in vacuum pressure, the lack of complete condensation of the steam released from the steam trap, and losses of heat into the atmosphere. There could also be some delays in the measurements of the temperatures by the probe. Errors could be minimized by insulating better, using an automated system in measuring the flows, or employing a more accurate pressure regulator.

The variable liquid level in the separator was another limitation, which varied the heating surface. The time it took the vacuum pump to react can also cause variations. Leaks can also occur. Better sealing, automation of the liquid level, or an instantaneous temperature and pressure sensor will help make the experiment more repeatable.

The assumptions of steady state conditions and heat losses can be considered only partially valid. For instance, despite attaining the state of stable conditions, there could be small variations resulting from the startup variation in vacuum levels. The assumption concerning the sugar solution acting like water does not account for the boiling-point elevation.

Definitive conclusions include that decreasing the pressure resulted in an increase in evaporator capacity and a decrease in the boiling-point elevation of the solution. The relationship of the economy with the pressure indicated the proper inverse relation of increased capacity with a marginal decrease in the economy with a deeper vacuum. The speculative conclusions would relate to how much sugar concentration or viscosity had influenced the value of the effective heat-transfer coefficient.

The findings confirm the theoretical foundation that vacuum operation enhances energy efficiency and product quality by reducing boiling point and inhibiting thermal degradation. At the same time, they indicate the realities of trade-offs in vacuum maintenance costs and inefficiencies from heat loss in small-scale processing. Future work could examine how feed concentration, multi-effect operation, or heat-transfer surface types could affect this process. It would be informative to compare experimental results with simulation models (such as Aspen Plus) to understand how well the models correlate with experimental data or how the overall heat-transfer coefficient could be adjusted in a non-ideal environment.

6. References (10 pts)

Begin your References section here. At least 4 references plus the lab manual are required. Make sure that the references are cited in the body of the report. The first reference in the report will be [1], the second reference [2], etc. Include a page break at the end of this section to start your Appendices.

[1] CHE 382 Unit Operations Laboratory, Evaporation Lab Manual, version 1, Fall 2025.

[2] R.H. Perry and D.W. Green, Eds., Perry's Chemical Engineer's Handbook, 9th edition. McGraw-Hill, 2018.

[3] McCabe, W. L., Smith, J. C., & Harriot, P. (2005). Unit operations of Chemical Engineering (7th ed.). McGraw-Hill Higher Education.

[4] Snee, P. T. (2024). Free Energy. Preston T. Snee. https://pchem.digital.uic.edu/

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7. Appendix I: Data Tabulation/Graphs (10 pts)

Trial Number	Vacuum Setting	Vacuum Pressure (inHg)	Time (s)	Steam Condensate Volume (mL)	Product Volume (mL)
1	1	-25	900	2730	1240
2	1	-25	900	2758	1251
3	2	-15	900	2321	690
4	2	-15	900	2352	712
5	3	-10	900	2202	446
6	3	-10	900	2170	430

Table 1: Raw Data (Part 1 of 2)

Trial Number	T1 (°C)	T2 (°C)	T3 (°C)	T4 (°C)	T5 (°C)	T6 (°C)	T7 (°C)	T8 (°C)	Heat Exchanger Pressure (kPa)
1	59	57	31	99	21	7	28	114	20
2	57	56	28	98	19	7	27	113	20
3	80	79	18	102	30	7	37	113	35
4	78	76	15	102	29	7	36	113	35
5	87	88	12	104	34	8	45	114	46
6	87	87	10	103	32	6	43	114	46

Table 2: Raw Data (Part 2 of 2)

Trial Number	T1, Heat Exchanger Outlet, WV (K)	T2, Condenser Inlet, WV (K)	T3, Condenser Outlet, WV, Product (K)	T4, Steam Trap Inlet, Steam (K)	T5, Steam Trap Outlet, CW (K)	T6, Steam Trap Inlet, CW (K)	T7, Steam Trap Outlet, Condensed Steam (K)	T8, Heat Exchanger Inlet, Steam (K)
1	332	330	304	372	294	280	301	387
2	330	329	301	371	292	280	300	386
3	353	352	291	375	303	280	310	386
4	351	349	288	375	302	280	309	386
5	360	361	285	377	307	281	318	387

6	360	360	283	376	305	279	316	387

Table 3: Temperatures in Kelvin

Trial Number	Vacuum Setting	Vacuum Pressure (in Hg)	Gauge Pressure (Torr/ mmHG)	Absolute Pressure (Torr/ mmHG)
1	1	-25	-635	125
2	1	-25	-635	125
3	2	-15	-381	379
4	2	-15	-381	379
5	3	-10	-254	506
6	3	-10	-254	506

Table 4: Pressure Calculations

Trial Number	Time (s)	Time (hr)	Steam Condensate Volume (mL)	Product Volume (mL)	Economy	Capacity (kg/hr)
1	900	0.2500	2730	1240	0.454	4.960
2	900	0.2500	2758	1251	0.454	5.004
3	900	0.2500	2321	690	0.297	2.760
4	900	0.2500	2352	712	0.303	2.848
5	900	0.2500	2202	446	0.203	1.784
6	900	0.2500	2170	430	0.198	1.720

Table 5: Economy and Capacity Calculations

Trial Number	T7, Steam Trap Outlet, Condensed Steam (K)	T8, Heat Exchanger Inlet, Steam (K)	T5, Steam Trap Outlet, CW (K)	T6, Steam Trap Inlet, CW (K)	Steam Flow Rate (kg/s)	qSTEAM (kJ/s)	qCW (kJ/s)	energy loss (kJ/s)	qHX (kJ/s)
1	301	387	294	280	3.03E-03	7.853	1.843	6.936	6.010
2	300	386	292	280	3.06E-03	7.940	1.580	6.995	6.360
3	310	386	303	280	2.58E-03	6.574	3.028	5.902	3.546
4	309	386	302	280	2.61E-03	6.673	2.897	5.979	3.776
5	318	387	307	281	2.45E-03	6.160	3.423	5.583	2.737
6	316	387	305	279	2.41E-03	6.091	3.42	5.501	2.668

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Table 6: Energy Balance Calculations

8. Appendix II: Error Analysis (10 pts)

<u>Qualitative error analysis</u>

Because this process design goal was to maintain a constant solution concentration by having a semi-continuous process feed of water, it could've resulted in a variety of sugar water concentrations throughout the experiment. The solution fluctuated between the indicated marks, which could've affected the evaporation process, influencing the boiling point, affecting the economy and capacity.

<u>Quantitative</u>

Absolute pressure (Torr/mm HG)	Average Econnomy(kg/hr) ± Stdev	Average Capacity ± Stdev
125	0.0412 ± 0.000044	4.9 ± 0.03
379	0.300 ± 0.0038	5.6 ± 0.06
506	0.200 ± 0.0031	1.8 ± 0.05

Table 7 Average Economy and Average capacity with standard deviation calculation

9. Appendix III: Sample Calculations (10 pts)

Sugar Solution:
The experiment calls for a 10 wt% sugar solution, and 2L will be used. Knowing that the density of water is approximately 1g/mL:

$$2000\, mL\, Water \cdot \frac{1\, g}{1\, mL} = 2000\, grams\, of\, water$$

$$w_1 = \frac{w_2 \cdot wtF}{1 - wtF}$$

Where w_2 is 2000 g, and wtF = 0.1

$$\frac{2000 \cdot 0.1}{1 - 0.1} = 222.2\, grams\, of\, sugar$$

Pressure:
For the first trial, the vacuum pressure was 25 inches of mercury:

$$-25 \in . Hg \frac{25.4\, torr}{1 \in . Hg} = -635\, torr$$

To get the absolute pressure, add 760 torr:

$$-635\, torr + 760\, torr = 125\, torr$$

Economy:
Economy is the ratio between kg of product/kg of steam used. Firstly, for the first trial, the mass of product is:

$$m_{product} = 1240\, mL \frac{1\, gram}{1\, mL} = 1240\, grams$$

Condensate Mass:

$$m_{condensate} = 2730\, mL \frac{1\, gram}{1\, mL} = 2730\, grams$$

Economy of Trial 1:

$$Economy = \frac{1240\, grams}{2730\, grams} = 0.45$$

The above calculations are done for all trials.

Capacity:
Capacity is the amount of product (condensate) collected per time (kg/hr). For trial one:

$$m_{product} = 1240\, mL \frac{1\, gram}{1\, mL} \frac{1\, kg}{1{,}000\, grams} = 1.24\, kilograms$$

Time:

$$900\, Seconds \frac{1\, hour}{3600\, Seconds} = 0.25\, hours$$

Capacity of Trial 1:

$$\frac{1.24\, kilograms}{0.25\, hours} = 4.96 \frac{kg}{hr}$$

This is repeated for all trials.

Energy Balances:

Assuming no energy loss, the energy balance on the heat exchanger is:

$$q_{HE} = q_{steam} - q_{CW}$$

Energy balance on steam:

$$q_{steam} = \dot{m}\left(\Delta T_{steam} C_{p,steam} + Hvap + \Delta T_{water} C_{p,water}\right)$$

To find $\dot{m}$:

$$\dot{m} = 2730\, mL \frac{1}{900\, s} \frac{1\, gram}{1\, mL} \frac{1\, kg}{1{,}000\, grams} = 0.003 \frac{kg}{s}$$

To find ΔT_{steam}:

$$\Delta T_{steam} = 387\, K - 373\, K = 14\, K$$

373K is the boiling point of water.

To find ΔT_{water}:

$$\Delta T_{water} = 373\, K - 301\, K = 72\, K$$

Heat capacity of steam is:

$$C_{p,steam} = 1.996 \frac{kJ}{kg\, K}$$

Heat capacity of water is:

$$C_{p,water} = 4.18 \frac{kJ}{kg\, K}$$

Heat of vaporization for water is:

$$Hvap = 2{,}260 \frac{kJ}{kg}$$

Plug in values:

$$q_{steam} = 0.003 \frac{kg}{s}\left(14\, K \cdot 1.996 \frac{kJ}{kg\, K} + 2{,}260 \frac{kJ}{kg} + 72\, K \cdot 4.18 \frac{kJ}{kg\, K}\right) = 7.85 \frac{kJ}{s}$$

Now, the energy balance for the cooling water in the trap:

$$q_{CW} = \dot{m}\, \Delta T_{CW} C_{p,CW}$$

$\dot{m}$ can be found by converting from volumetric flow rate:

$$\dot{m} = 0.2385\, gpm \frac{0.063 \frac{kg}{s}}{1\, gpm} = 0.015 \frac{kg}{s}$$

Heat capacity of the cooling water is:

$$C_{p,CW} = 4.18 \frac{kJ}{kg\, K}$$

ΔT_{CW} is found by taking the difference between the temperature of the cooling water entering the steam trap and the temperature leaving the stream trap:

$$\Delta T_{CW} = 294\, K - 280\, K = 14\, K$$

Plug in values:

$$q_{CW} = 0.015 \frac{kg}{s} \cdot 14\, K \cdot 4.18 \frac{kJ}{kg\, K} = 1.84 \frac{kJ}{s}$$

Energy balance on heat exchanger:

$$q_{HE} = 7.85 \frac{kJ}{s} - 1.84 \frac{kJ}{s} = 6.93 \frac{kJ}{s}$$

These calculations are done for all trials.

To find the energy loss in the heat exchanger:

$$Energy\ Loss\, of\ HE = \dot{m}_{steam} \cdot Hvap + Capacity \cdot C_{p,water} \cdot \Delta T$$

Heat of vaporization for water:

$$Hvap = 2{,}260 \frac{kJ}{kg}$$

Trial 1 capacity:

$$4.96 \frac{kg}{hr}$$

Heat capacity of water is:

$$C_{p,water} = 4.18 \frac{kJ}{kg\, K}$$

ΔT is found by taking the difference between the temperature of the cooling water that is entering the steam trap and the temperature that is leaving the steam trap.

$$\Delta T = 294\, K - 280\, K = 14\, K$$

To find $\dot{m}_{steam}$:

$$\dot{m}_{steam}=2730\,mL\,\frac{1}{900\,s}\,\frac{1\,gram}{1\,mL}\,\frac{1\,kg}{1{,}000\,grams}=0.003\,\frac{kg}{s}$$

Plug in values:

$$Energy\ Loss\,of\ HE=0.003\,\frac{kg}{s}\cdot 2{,}260\,\frac{kJ}{kg}+4.96\,\frac{kg}{hr}\,\frac{1\,hr}{3600\,s}\cdot 4.18\,\frac{kJ}{kg\,K}\cdot 14\,K=7.0\,\frac{kJ}{s}$$

These calculations are done for all trials.

10. Appendix IV: Individual Team Contributions

NAME: Aaron Casas

	TIME (HOURS)	**DESCRIPTION**
Operator (Both Lab Days)	6	In lab operating hours
Pre-Lab Editing	1	Finalized the safety portion of the lab
Experimental Objective (Prelab)	1	Formulated steps needed to be taken to complete the lab
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	1	Made safety guidelines
Final Lab Editing	3	Checked for completion of each section.
Executive Summary (final lab)	0	
Introduction (Final lab)	1	
Theory (Final Lab)	0	
Results (final lab)	1	
Discussion (final lab)	0	
References (final	2	Referenced sources as

and/or prelab)		well as in-text citations
Data Tabulation/Graphs (final)	2	Formulated the tables.
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	2	Did sample calculations on excel.
Power Point Presentation	1	Added information to slides.
Total	21	

NAME: Hussein Rezk

	TIME (HOURS)	**DESCRIPTION**
Operator (Both Lab Days)	8	Read and conducted an experiment
Pre-Lab Editing	3	Assisted in all parts
Experimental Objective (Prelab)	2	Assisted in objective creation
Apparatus (prelab)	1	Assisted in the assembly of the apparatus
Materials & Supplies (prelab)	1	Created a list of materials and supplies
Procedure (prelab)	2	Helped create a procedure for the experiment report
Project Safety Evaluation (prelab)	1	Made safety guidelines
Final Lab Editing	3	Made safety guidelines

Executive Summary (final lab)	2	Assisted in the formation of the summary
Introduction (Final lab)	0	
Theory (Final Lab)	3	Found valuable literature and theory to perform an analytical comparison
Results (final lab)	0	
Discussion (final lab)	4	Analyzed correlations, theory linearity, and the correctness of experiment results
References (final and/or prelab)	1	Tabulated references
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	1	Conducted analysis on data
Sample Calculations (final Lab)	0	
Power Point Presentation	2	Created in collaboration with group
Total	33	

NAME: Adil Feratovic

	TIME (HOURS)	**DESCRIPTION**
Operator (Both Lab Days)	8	Assisted in conduction of lab
Pre-Lab Editing	2	Made objective portion
Experimental Objective (Prelab)	1	Created the objective
Apparatus (prelab)	0	

Materials & Supplies (prelab)	0	
Procedure (prelab)	2	Created a procedure for the experiment report
Project Safety Evaluation (prelab)	0	
Final Lab Editing	4	Worked on formatting, sig figs
Executive Summary (final lab)	1	Worked on executive summary
Introduction (Final lab)	2	Worked on introduction
Theory (Final Lab)	3	Worked on theory
Results (final lab)	5	Worked on results
Discussion (final lab)	4	Worked on discussion
References (final and/or prelab)	1	Added references
Data Tabulation/Graphs (final)	2	Worked on calculations
Error Analysis (final lab)	2	Did an error analysis
Sample Calculations (final Lab)	2	Wrote out the sample calculations
Power Point Presentation	1	Worked on presentation
Total	38	

NAME: Anna Cai

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Conducted experiment
Pre-Lab Editing	3	Edited grammar
Experimental Objective (Prelab)	0	
Apparatus (prelab)	2	Took photos of and wrote descriptions for the apparatus
Materials & Supplies (prelab)	1	Took into account materials and supplies needed
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	3	Edited and finalized grammar
Executive Summary (final lab)	1	Assisted in writing the executive summary
Introduction (Final lab)	0	
Theory (Final Lab)		
Results (final lab)	2	Helped calculate results
Discussion (final lab)	2	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	2	Wrote out sample calculations
Power Point Presentation	1	Finalized powerpoints
Total	25	

REFERENCES for this Report Template Document:

1. Final Report Template Fall 2009, Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL.
2. CHE4162 Syllabus Spring 2010, Department of Chemical Engineering, Louisiana State University, Baton Rouge, LA.
3. [1] CHE 382 Unit Operations Laboratory, Evaporation Lab Manual, version 1, Fall 2023.
4. [2] R.H. Perry and D.W. Green, Eds., Perry’s Chemical Engineer’s Handbook, 9th edition. McGraw-Hill, 2018.
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